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Lab1

8 exercises

Note: Shared utils code is shown at the very bottom of this report (after all exercises), in both HTML and exported PDF.

1 Ex1

1 file

EXERCISE DESCRIPTION

Write a C++ program to ask information from a student such as name, gender, major, age then display the greeting message on screen. Say he/she is eligible to vote the election if the age is at least 18. Hi, Mr. name! your age is age year old and you learn major!   : for male (M) Hi, Mrs. name!, you are age year old and you are majoring in major : for female (F)

For either Male or Female:
You can vote.   If age is at least 18.
You can not vote Otherwise.

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>

using namespace std;
struct Student {
    string name;
    char gender;
    string major;
    int age;
};

int main() {
    Student student;
    student.name = getString("Enter name: ");
    student.gender = getGender("Enter gender (M/F): ");
    student.major = getString("Enter major: ");
    student.age = getInteger("Enter age: ");

    cout << "Hi, " << ((student.gender == 'M') ? "Mr" : "Mrs") << ". " << student.name
         << "! your age is " << student.age << " year old and you learn " << student.major << "!"
         << endl;
    return 0;
}
```

hey.csv OUTPUT

unt	wce	wec	we	we
1	12	12	31	13

bbg.txt OUTPUT

```
cunt
```

output.txt OUTPUT

```
Enter name: Hengrith
Enter gender (M/F): M
Enter major: SE
Enter age: 20
Hi, Mr. Hengrith! your age is 20 year old and you learn SE!
```

2 Ex2

1 file

EXERCISE DESCRIPTION

Write a C++ program to check whether an input character is a vowel or a consonant. Hint: Use ASCII code to test condition. ▪ Uppercase letters from: 65 to 90, lowercase letters from 97 to 122 ▪ Vowel and its ASCII code: a = 97, e = 101, i = 105, o = 111, u = 117

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <cctype>
#include <iostream>

using namespace std;

int main() {
    char input = getChar("Enter a character: ");

    input = tolower(input);

    bool isVowel = (input == 97 || input == 101 || input == 105 || input == 111 || input == 117);

    if (isVowel) {
        cout << "Vowel." << endl;
    } else {
        cout << "Not a vowel." << endl;
    }

    return 0;
}
```

output.txt · section 1 OUTPUT

```
Enter a character: a
Vowel.
```

output.txt · section 2 OUTPUT

```
Enter a character: b
Not a vowel.
```

3 Ex3

1 file

EXERCISE DESCRIPTION

Write a C++ program to compute tax salary of a person. The program asks for name, gender and salary of a person and tell him/her how much tax he/she is required to pay. The tax is computed based on the rules below:

- For males
  - Salary more than 1000 USD, pay tax 9.5%
  - Salary 500 – 1000 USD, pay tax 7%
  - Salary 300 – 500 USD, pay tax 5%
  - Salary less than 300, no need to pay tax
- For females
  - Salary more than 1000 USD, pay tax 8%
  - Salary 500 – 1000 USD, pay tax 6.5%
  - Salary 300 – 500 USD, pay tax 3.5%
  - Salary less than 300, no need to pay tax

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>

using namespace std;

int main() {
    string name = getString("Enter name: ");
    char gender = getGender("Enter gender (M/F): ");
    double salary = getDouble("Enter salary: ");

    double taxRate = 0;

    if (toupper(gender) == 'M') {
        if (salary > 1000) taxRate = 0.095;
        else if (salary >= 500) taxRate = 0.07;
        else if (salary >= 300) taxRate = 0.05;

    } else {
        if (salary > 1000) taxRate = 0.08;
        else if (salary >= 500) taxRate = 0.065;
        else if (salary >= 300) taxRate = 0.035;
    }

    double tax = salary * taxRate;
    cout << endl;
    cout << "Name: " << name << endl;
    cout << "Salary: $" << salary << endl;

    if (tax == 0) {
        cout << "No tax required." << endl;
    } else {
        cout << "Tax rate: " << (taxRate * 100) << "%" << endl;
        cout << "Tax to pay: $" << tax << endl;
    }

    return 0;
}
```

output.txt · section 1 OUTPUT

```
Enter name: Bob
Enter gender (M/F): m
Enter salary: 299

Name: Bob
Salary: $299
No tax required.
```

output.txt · section 2 OUTPUT

```
Enter name: gyaru
Enter gender (M/F): F
Enter salary: 300

Name: gyaru
Salary: $300
Tax rate: 3.5%
Tax to pay: $10.5
```

4 Ex4

1 file

EXERCISE DESCRIPTION

Create a C++ program that can play a game Rock-Paper-Scissor. The game plays between the user against the computer.

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>
#include <random>

using namespace std;

int determineWin(int user, int computer) {
    if (user == computer) return 0;

    if ((user == 1 && computer == 2) ||
        (user == 2 && computer == 3) ||
        (user == 3 && computer == 1))
        return 1;

    return -1;
}

int main() {

    random_device rd;
    mt19937 gen(rd());
    uniform_int_distribution<> dist(1, 3);

    int option;
    int computerChoice;
    int result;

    while (true) {
        cout << "=== Rock-Paper-Scissor ===" << endl;
        cout << "1. Paper" << endl;
        cout << "2. Rock" << endl;
        cout << "3. Scissor" << endl;
        cout << "0. Exit" << endl;

        option = getOption("Option: ", 3);
        if (option == 0) break;

        computerChoice = dist(gen);

        result = determineWin(option, computerChoice);

        if (result == 0) cout << "Draw\n";
        else if (result == 1) cout << "You win!\n";
        else cout << "You lose!\n";

        cout << endl;
    }

    return 0;
}
```

output.txt · section 1 OUTPUT

```
=== Rock-Paper-Scissor ===
1. Paper
2. Rock
3. Scissor
0. Exit
Option: 1
You win!
```

output.txt · section 2 OUTPUT

```
=== Rock-Paper-Scissor ===
1. Paper
2. Rock
3. Scissor
0. Exit
Option: 1
Draw
```

output.txt · section 3 OUTPUT

```
=== Rock-Paper-Scissor ===
1. Paper
2. Rock
3. Scissor
0. Exit
Option: 1
You lose!
```

output.txt · section 4 OUTPUT

```
=== Rock-Paper-Scissor ===
1. Paper
2. Rock
3. Scissor
0. Exit
Option: 0
```

5 Ex5

1 file

EXERCISE DESCRIPTION

Write a C++ program to convert a minute to a time format which consists of hour, minute, and second (h:m:s). A user is required to input a minute.

- Hours = minutes / 60
- Remainderminutes = minutes % 60
- Seconds = remainderminutes * 60

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>

using namespace std;

int main() {
    int minutes = getInteger("Enter minutes: ");

    int hours = minutes / 60;            // get hours
    int remainderMinutes = minutes % 60; // get remaining minutes
    int seconds = remainderMinutes * 60; // convert remaining minutes to seconds

    cout << hours << "h : " << remainderMinutes << "m : " << seconds << "s" << endl;

    return 0;
}
```

output.txt OUTPUT

```
Enter minutes: 1039
17h : 19m : 1140s
```

6 Ex6

1 file

EXERCISE DESCRIPTION

Write a C++ program to find the summation of numbers from 1 to n except number 10 and 30, where n is a number input by a user and n should be greater than 50.

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>

using namespace std;

int main() {
    int n;

    while (true) {
        n = getInteger("Enter a number greater than 50: ");
        if (n > 50) break;
        cout << "Invalid input. Please enter a number greater than 50." << endl;
    }

    int sum = 0;
    for (int i = 1; i <= n; i++) {
        if (i == 10 || i == 30) continue;
        sum += i;
    }

    cout << "Summation from 1 to " << n << " (except 10 and 30) = " << sum << endl;

    return 0;
}
```

output.txt OUTPUT

```
Enter a number greater than 50: 50
Invalid input. Please enter a number greater than 50.
Enter a number greater than 50: 51
Summation from 1 to 51 (except 10 and 30) = 1286
```

7 Ex7

1 file

EXERCISE DESCRIPTION

Write a C++ program to create a new data structure for storing info of book (book ID, book ISBN, book title, published year, author names and price ($)). Each book could have more than one
author. To do:

- Create an array that can store 5 books' info.
- Create a function to display a book info based on ISBN. This function takes a parameter which is an ISBN of a book.

void displayBookByISBN(Book books[], int size, string isbn){.... }

- Create a function to display information of all books.

void displayAllBooks(Book books[], int size) { ... }

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <iostream>

using namespace std;

struct Book {
    int id;
    string isbn;
    string title;
    int publishedYear;
    string authors[5];
    int authorCount;
    double price;
};

void displayBook(Book book) {
    cout << "ID:" << book.id << endl;
    cout << "ISBN: " << book.isbn << endl;
    cout << "Title: " << book.title << endl;
    cout << "Published Year: " << book.publishedYear << endl;
    cout << "Price:$ " << book.price << endl;
    cout << "Authors: ";
    for (int i = 0; i < book.authorCount; i++) {
        cout << book.authors[i];
        if (i < book.authorCount - 1) cout << ", ";
    }
    cout << endl
         << endl;
}

void displayBookByISBN(Book books[], int size, string isbn) {
    for (int i = 0; i < size; i++) {
        if (books[i].isbn == isbn) {
            displayBook(books[i]);
            return;
        }
    }
    cout << "Book with ISBN " << isbn << " not found." << endl;
}

void displayAllBooks(Book books[], int size) {
    for (int i = 0; i < size; i++) {
        displayBook(books[i]);
    }
}

int main() {
    Book books[5] = {
        {1, "978-0-13-468599-1", "The C++ Programming Language", 2013, {"Bjarne Stroustrup"}, 1, 59.99},
        {2, "978-0-13-235088-4", "Clean Code", 2008, {"Robert C. Martin"}, 1, 35.99},
        {3, "978-0-20-163361-0", "The Pragmatic Programmer", 1999, {"Andrew Hunt", "David Thomas"}, 2, 49.99},
        {4, "978-0-59-651798-1", "JavaScript: The Good Parts", 2008, {"Douglas Crockford"}, 1, 29.99},
        {5, "978-0-13-110362-7", "The C Programming Language", 1988, {"Brian Kernighan", "Dennis Ritchie"}, 2,
39.99}};

    displayAllBooks(books, 5);

    cout << "\n--- Search by ISBN ---" << endl;
    string isbn = getString("Enter ISBN: ");
    displayBookByISBN(books, 5, isbn);

    return 0;
}
```

output.txt OUTPUT

```
ID:1
ISBN: 978-0-13-468599-1
Title: The C++ Programming Language
Published Year: 2013
Price:$ 59.99
Authors: Bjarne Stroustrup

ID:2
ISBN: 978-0-13-235088-4
Title: Clean Code
Published Year: 2008
Price:$ 35.99
Authors: Robert C. Martin

ID:3
ISBN: 978-0-20-163361-0
Title: The Pragmatic Programmer
Published Year: 1999
Price:$ 49.99
Authors: Andrew Hunt, David Thomas

ID:4
ISBN: 978-0-59-651798-1
Title: JavaScript: The Good Parts
Published Year: 2008
Price:$ 29.99
Authors: Douglas Crockford

ID:5
ISBN: 978-0-13-110362-7
Title: The C Programming Language
Published Year: 1988
Price:$ 39.99
Authors: Brian Kernighan, Dennis Ritchie


--- Search by ISBN ---
Enter ISBN: 978-0-13-110362-7
ID:5
ISBN: 978-0-13-110362-7
Title: The C Programming Language
Published Year: 1988
Price:$ 39.99
Authors: Brian Kernighan, Dennis Ritchie
```

8 Ex8

1 file

EXERCISE DESCRIPTION

```
Create 5 functions to:
i)   convert temperature Celsius to Fahrenheit: Fahrenheit = (Celsius * 9.0 / 5.0) + 32
ii)  convert temperature Fahrenheit to Celsius: Celsius = (Fahrenheit - 32) * 5.0 / 9.0
iii) find root of quadratic equation ax2+bx+c=0.
iv)  compute BMI of a person's weight and height and tell whether he/she is
overweight, underweight and other terms according to BMI list:
     - bmi = weight / (height * height)
     - bmi < 18.5, underweight
     - bmi < 25,   normal weight
     - bmi < 30,   overweight
     - else,       obese
v)   sum numbers from 1 to n except those numbers that are divisible by 3, n is a parameter.

Design a menu program to demonstrate these 5 functions.
```

Main.cpp MAIN

```cpp
#include "../../utils/input.cpp"
#include <cmath>
#include <iostream>

using namespace std;

void celsiusToFahrenheit() {
    double celsius = getDouble("Enter Celsius: ");
    double fahrenheit = (celsius * 9.0 / 5.0) + 32;
    cout << celsius << " celsius = " << fahrenheit << " fahrenheit" << endl;
}

void fahrenheitToCelsius() {
    double fahrenheit = getDouble("Enter Fahrenheit: ");
    double celsius = (fahrenheit - 32) * 5.0 / 9.0;
    cout << fahrenheit << " fahrenheit = " << celsius << " celsius" << endl;
}

void quadraticRoots() {
    double a = getDouble("Enter a: ");
    double b = getDouble("Enter b: ");
    double c = getDouble("Enter c: ");

    double discriminant = (b * b) - (4 * a * c);

    if (discriminant > 0) {
        double root1 = (-b + sqrt(discriminant)) / (2 * a);
        double root2 = (-b - sqrt(discriminant)) / (2 * a);
        cout << "Two roots: " << root1 << " and " << root2 << endl;
    } else if (discriminant == 0) {
        double root = -b / (2 * a);
        cout << "One root: " << root << endl;
    } else {
        cout << "No real roots." << endl;
    }
}

void bmiCalculator() {
    double weight = getDouble("Enter weight (kg): ");
    double height = getDouble("Enter height (m): ");

    double bmi = weight / (height * height);

    cout << "BMI: " << bmi << " - ";

    if (bmi < 18.5) cout << "Underweight" << endl;
    else if (bmi < 25) cout << "Normal weight" << endl;
    else if (bmi < 30) cout << "Overweight" << endl;
    else cout << "Obese" << endl;
}

void sumExceptDivisibleBy3() {
    int n = getInteger("Enter n: ");
    int sum = 0;

    for (int i = 1; i <= n; i++) {
        if (i % 3 == 0) continue;
        sum += i;
    }

    cout << "Sum from 1 to " << n << " (except divisible by 3) = " << sum << endl;
}

int main() {
    while (true) {
        cout << endl;
        cout << "1. Celsius to Fahrenheit" << endl;
        cout << "2. Fahrenheit to Celsius" << endl;
        cout << "3. Quadratic Equation Roots" << endl;
        cout << "4. BMI Calculator" << endl;
        cout << "5. Sum 1 to n (except divisible by 3)" << endl;
        cout << "0. Exit" << endl;

        int option = getOption("Enter option: ", 5);

        if (option == 0) break;
        else if (option == 1) celsiusToFahrenheit();
        else if (option == 2) fahrenheitToCelsius();
        else if (option == 3) quadraticRoots();
        else if (option == 4) bmiCalculator();
        else if (option == 5) sumExceptDivisibleBy3();
    }

    return 0;
}
```

output.txt · section 1 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 1
Enter Celsius: 50
50 celsius = 122 fahrenheit
```

output.txt · section 2 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 2
Enter Fahrenheit: 122
122 fahrenheit = 50 celsius
```

output.txt · section 3 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 3
Enter a: 3
Enter b: -5
Enter c: -8
Two roots: 2.66667 and -1
```

output.txt · section 4 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 4
Enter weight (kg): 49.5
Enter height (m): 1.65
BMI: 18.1818 - Underweight
```

output.txt · section 5 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 5
Enter n: 67
Sum from 1 to 67 (except divisible by 3) = 1519
```

output.txt · section 6 OUTPUT

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Quadratic Equation Roots
4. BMI Calculator
5. Sum 1 to n (except divisible by 3)
0. Exit
Enter option: 0
```

SHARED UTILITIES

Source: Lab/utils

input.cpp .cpp

```cpp
#include <iostream>
#include <sstream>
#include <string>
#include <cctype>

using namespace std;

template <typename T>
bool tryParseExact(const string& line, T& out) {
    stringstream stream(line); // treat the string as a stream to read from

    stream >> out; // try to read the value
    if (stream.fail()) {
        return false;
    }

    // Ignore trailing spaces, then require end-of-input.
    stream >> ws;
    return stream.eof();
}

int getInteger(string comment) {
    string line;
    int x;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (tryParseExact(line, x)) {
            return x;
        }

        cout << "Invalid input. Please enter a whole number." << endl;
    }
}

float getFloat(string comment) {
    string line;
    float x;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (tryParseExact(line, x)) {
            return x;
        }

        cout << "Invalid input. Please enter a float number." << endl;
    }
}

double getDouble(string comment) {
    string line;
    double x;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (tryParseExact(line, x)) {
            return x;
        }

        cout << "Invalid input. Please enter a double number." << endl;
    }
}

string getString(string comment) {
    string line;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (!line.empty()) {
            return line;
        }

        cout << "Invalid input. Please enter a non-empty string." << endl;
    }
}

char getGender(string comment) {
    string line;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (line.size() == 1) {
            char c = static_cast<char>(toupper(static_cast<unsigned char>(line[0])));
            if (c == 'M' || c == 'F') {
                return c;
            }
        }

        cout << "Invalid input. Please enter M or F." << endl;
    }
}

char getChar(string comment) {
    string line;

    while (true) {
        cout << comment;
        getline(cin, line);

        if (line.size() == 1) {
            return line[0];
        }

        cout << "Invalid input. Please enter exactly one character." << endl;
    }
}

int getOption(string comment, int maxOption) {
    if (maxOption < 0) {
        return 0;
    }

    while (true) {
        int option = getInteger(comment);
        if (option >= 0 && option <= maxOption) {
            return option;
        }

        cout << "Invalid option. Please enter a number from 0 to "
             << maxOption << "." << endl;
    }
}
```